

Emergent Geometric Phase Structure in Minimal Relational Networks

Snek Pas

Independent Researcher

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The emergence of large-scale geometric organization from fundamentally discrete, non-spatial relational dynamics remains a central challenge in complex systems and pre-geometric models of quantum gravity. We present a minimal generative network ensemble driven by stochastic edge toggling under a purely local, quadratic degree-regularization penalty, without any externally imposed embedding space. Sweeps of the regularization strength reveal that the network self-organizes into discrete coordination plateaus that sharpen with system size, organizing structural and dynamical observables into phase-like families. Finite-size scaling of plateau breakout points demonstrates convergence to a non-zero thermodynamic critical boundary, establishing the phenomenon as a genuine macroscopic phase transition. Within plateau regimes, we observe dimensional decoupling in which the spectral dimension (d_S) stabilizes while the Hausdorff dimension (d_H) exhibits small-world divergence, indicating separation between diffusion and geodesic volume geometry. Varying a triadic locality bias systematically tunes diffusion geometry without disrupting the underlying coordination phase structure. These results demonstrate that stable, tunable diffusion geometry can arise generically as a collective coordination phase driven solely by relational regularization dynamics.

I. INTRODUCTION

How persistent large-scale organization emerges from simple microscopic rules remains a central question across physics, biology, and complex systems. Networks provide a minimal representation of relational structure, yet most models either assume fixed topology or impose geometry externally. Here we explore a minimal generative framework in which geometry-like signatures arise from relational dynamics alone.

In many systems, distinct notions of dimension need not coincide: shortest-path scaling can indicate small-world behavior while diffusion remains effectively finite-dimensional. We demonstrate that this dimensional decoupling can be robust and controllable in an algorithmically generated network ensemble governed primarily by degree regularization and locality bias.

II. MODEL

We consider an undirected simple graph $G(V, E)$ with N nodes. The system evolves via stochastic toggling of candidate edges (u, v) , accepted by a Metropolis criterion using an energy change ΔE that includes:

(i) a baseline edge cost λ , (ii) a quadratic degree regularization term of strength μ , and optionally (iii) a triadic/triangle term (set to zero in the minimal model).

Candidate edges are proposed by a mixture of:

- *Local peer-introduction* (probability p_{triadic}): choose a node and attempt closure between two of its neighbors;
- *Global proposal* (probability $1 - p_{\text{triadic}}$): choose a random node pair.

An annealing schedule decreases effective temperature T toward zero so that the system relaxes into stable plateau regimes.

III. OBSERVABLES

We track:

- mean degree $\langle k \rangle = 2|E|/N$,
- transitivity (global clustering) C ,
- assortativity r ,
- Hausdorff dimension d_H estimated from volume growth of graph-distance balls,
- spectral dimension d_S estimated from random-walk return probability scaling.

IV. DEGREE-PENALTY SWEEPS AND PLATEAU PHASES

Sweeps over μ reveal that the induced mean degree $\langle k \rangle(\mu)$ self-organizes into discrete plateaus that sharpen with increasing system size. Across these plateaus, structural and dynamical observables collapse into families, indicating Hamiltonian-controlled phase structure.

V. FINITE-SIZE SCALING AND THE THERMODYNAMIC LIMIT

To confirm that the observed geometric organization represents a true phase transition rather than a finite-size computational artifact, we analyzed the critical degree penalty $\mu_c(N)$ at which the system breaks out of its stable plateau regime. As shown in Fig. 2, smaller networks destabilize at higher penalty values.

By fitting these critical points to a standard finite-size scaling power law, $\mu_c(N) = \mu_c(\infty) + A \cdot N^{-\theta}$, we extrapolate the thermodynamic limit ($1/N \rightarrow 0$). The scaling converges to a non-zero critical boundary of

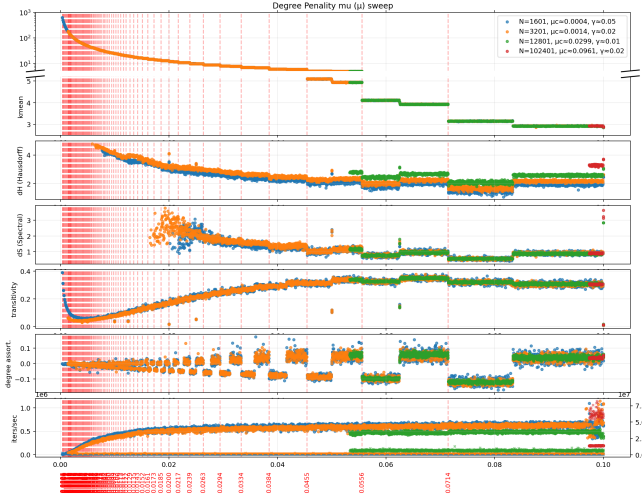


FIG. 1: Degree-penalty μ sweep across system sizes. Discrete $\langle k \rangle$ plateaus organize families of observables (e.g., d_S , d_H , clustering, assortativity) into phase-like regimes.

$\mu_c(\infty) \approx 0.04155$, distinguishing the observed coordination plateaus from finite-size artifacts that would vanish for large systems. Instead, the results indicate a genuine collective transition in relational organization: below the critical regularization strength, stochastic edge dynamics remain insufficient to stabilize global coordination, whereas above this threshold the network self-organizes into persistent coordination regimes supporting stable diffusion geometry. In this sense, the degree-regularization parameter acts as an effective coordination stiffness whose finite critical value establishes emergent geometric organization as a macroscopic phase phenomenon.

VI. DIMENSIONAL DECOUPLING

Within plateau regimes, d_S stabilizes to near-constant values with weak size dependence, while d_H continues to exhibit small-world divergence. This demonstrates robust decoupling of diffusion geometry from geodesic volume-growth geometry.

VII. LOCALITY CONTROL VIA p_{triadic}

Varying p_{triadic} leaves the $\langle k \rangle$ plateau structure largely invariant but systematically shifts diffusion geometry as quantified by d_S . Increased triadic locality increases motif redundancy and enhances return probability, lowering d_S , while increased global proposals enhance mixing and raise d_S .

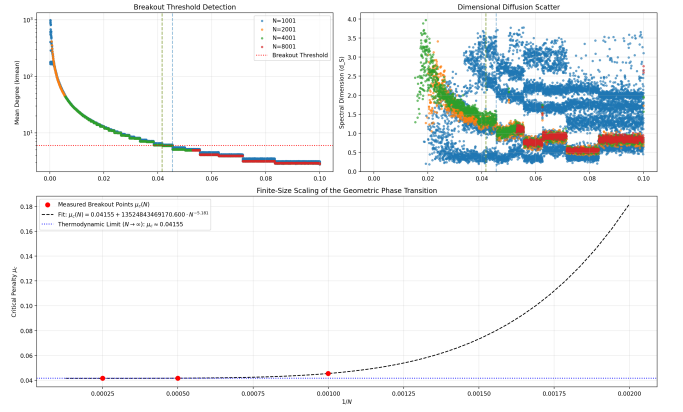


FIG. 2: Finite-size scaling and dimensional decoupling at the geometric phase transition. (Top Left) Mean degree $\langle k \rangle$ as a function of the degree regularization penalty μ for system sizes $N \in \{1001, 2001, 4001, 8001\}$. The structural breakout point $\mu_c(N)$ is defined by the threshold $\langle k \rangle > 6.0$ (red dotted line). (Top Right) The corresponding spectral dimension d_S scatters sharply at reminder of the structural breakout points (vertical dashed lines). (Bottom) Finite-size scaling of the critical breakout points $\mu_c(N)$ plotted against $1/N$. The extrapolation to a non-zero thermodynamic limit identifies μ as an effective coordination stiffness separating disordered relational dynamics from an emergent coordination phase supporting stable diffusion geometry.

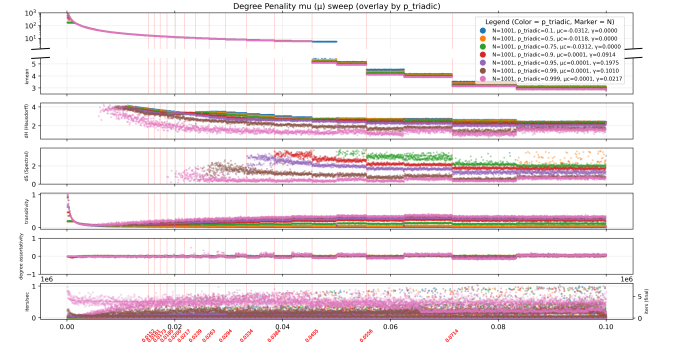


FIG. 3: μ sweep with multiple p_{triadic} values overlaid. Mean-degree plateaus persist across p_{triadic} (Hamiltonian-controlled), while d_S shifts within each plateau (locality-controlled diffusion geometry).

VIII. EMERGENCE TRAJECTORY: FROM SPARSE FLUCTUATIONS TO STABLE FABRIC

To illustrate the dynamics of emergence, Fig. 4 shows a sequence of snapshots (fixed N and seed) as connectivity increases from near-empty initial conditions into a stable plateau regime. Even before full densification, spectral/embedding structure nucleates and consolidates into a coherent bulk.

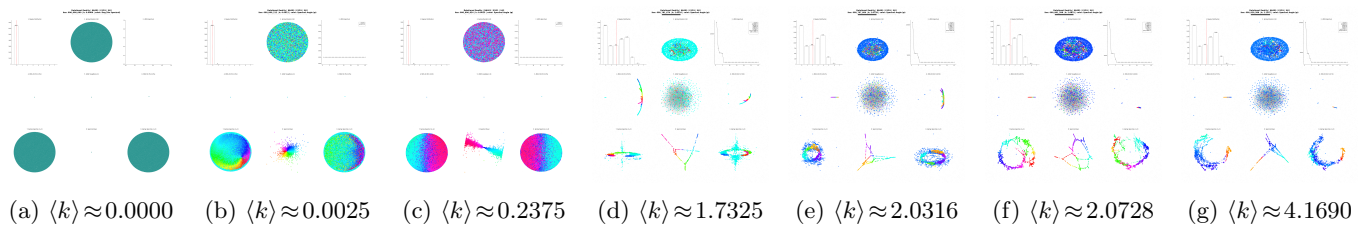


FIG. 4: Emergence sequence for a large- N run showing the transition from sparse fluctuations to a stabilized high-connectivity regime. Connectivity increases monotonically and the system condenses into a stable fabric-like macrostate.

IX. MINIMALITY AND ROBUSTNESS

Progressive removal of model components indicates that plateau formation and dimensional signatures persist under a minimal Hamiltonian dominated by degree regularization plus a locality-controlled proposal distribution. This suggests that geometric phase behavior arises generically from relational regularization dynamics rather than from fine-tuned microterms.

X. DISCUSSION

The observed plateau organization, dimensional decoupling, and locality-controlled diffusion geometry support an interpretation of emergent geometry as a collective coordination phase in relational systems. Finite-size scaling demonstrating a non-zero critical regularization boundary indicates that these structures are not finite-size artifacts but reflect macroscopic ordering within the relational ensemble. Below the critical coordination stiffness, fluctuations dominate and connectivity remains weakly organized, whereas above this threshold coordination pressure stabilizes discrete connectivity shells supporting persistent diffusion geometry.

This transition bears qualitative resemblance to ordering phenomena such as crystallization or jamming, where local interaction constraints collectively stabilize global structure. In contrast to conventional systems, no spatial embedding or metric constraints are imposed here; coordination regularization alone generates discrete connectivity regimes analogous to coordination numbers in atomic packings. The results therefore suggest that ge-

ometric organization can arise as a coordination-driven collective phase even in purely relational settings.

More broadly, the separation between Hamiltonian-controlled coordination structure and locality-controlled diffusion geometry indicates that multiple geometric characteristics can emerge and be tuned independently within the same relational phase. While the present work does not attempt to derive continuum gravitational dynamics, it demonstrates that geometry-like signatures can arise as collective organization phenomena without embedding assumptions.

Future work will include bidirectional sweeps to quantify hysteresis windows, alternative coordination regularizers to probe universality, refined estimation of transition widths, and systematic identification of minimal order parameters beyond $\langle k \rangle$ (e.g., triadic saturation and spectral gap) for sharper phase classification.

XI. CONCLUSION

Minimal relational network dynamics exhibit robust geometric phase structure: discrete coordination plateaus, stabilized diffusion dimensionality within plateaus, and independent control of diffusion geometry via locality bias. The extrapolation of a non-zero thermodynamic limit confirms these structures as genuine phase phenomena. These results motivate further study of emergent geometry in minimal relational ensembles and provide a reproducible computational laboratory for phase-structured network geometry.

These findings support the interpretation of geometric organization as an ordering transition in coordination space rather than in physical space.